

SOUTH WEST CAPE, New Zealand

WMO: 939090

Lat: 47.277S

Lon: 167.463E

Elev: 336

StdP: 14.52

Time Zone: 12.00 (NZL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.4	38.9	28.9	23.1	41.4	30.8	25.3	42.1	58.5	50.0	54.8	49.4	23.6	230	1.069

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.6	67.0	59.4	64.6	58.2	62.8	57.2	61.3	65.0	59.7	63.1	58.3	61.6	19.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.7	77.4	62.7	58.1	72.9	61.3	56.7	69.3	60.0	27.5	64.7	26.4	63.3	25.5	61.7	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	52.7	48.1	35.1	73.8	1.4	3.8	34.0	76.5	33.2	78.7	32.4	80.8	31.4	83.5
			32.8	63.8	1.4	2.3	31.8	65.4	31.0	66.8	30.2	68.1	29.2	69.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	55.0	55.4	54.3	51.7	49.4	47.0	46.0	46.6	48.0	48.7	50.3	53.7	(6)
	DBStd	4.88	3.77	3.92	3.87	3.73	4.00	3.33	3.22	3.38	3.54	3.29	3.59	3.92	(7)
	HDD50	620	2	2	5	22	59	100	129	112	79	64	37	9	(8)
	HDD65	5307	312	270	333	400	484	541	589	570	510	506	440	352	(9)
	CDD50	792	156	153	137	72	41	9	5	7	19	24	47	122	(10)
	CDD65	2	1	1	0	0	0	0	0	0	0	0	0	0	(11)
	CDH74	4	1	2	1	0	0	0	0	0	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14) Wind	WSAvg	23.9	24.9	22.7	22.9	24.0	24.7	24.1	22.9	22.3	24.4	25.7	25.3	23.2	(14)
(15) Precipitation	PrecAvg	52.7	4.3	3.9	4.5	4.5	5.0	4.5	4.4	4.1	4.2	4.6	4.3	4.3	(15)
	PrecMax	67.0	7.0	7.6	11.7	10.1	8.0	7.5	9.8	6.2	6.9	7.4	7.3	7.1	(16)
	PrecMin	45.7	1.4	1.3	1.9	1.9	1.2	1.8	2.5	3.1	1.8	1.8	1.3	2.2	(17)
	PrecStd	5.2	1.5	1.7	2.3	1.9	1.7	1.5	1.7	0.7	1.5	1.5	1.5	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	71.7	68.1	64.5	61.4	57.1	55.6	56.6	60.4	61.6	64.1	69.2	(19)
		MCWB	61.3	62.0	60.0	56.6	52.8	49.5	48.8	48.6	51.1	52.8	56.4	60.2	(20)
	2%	DB	66.1	66.6	64.7	61.1	58.8	54.1	53.6	54.1	56.9	58.2	60.6	64.6	(21)
		MCWB	59.7	59.5	58.6	54.7	52.8	50.1	48.7	47.9	50.1	51.9	54.3	57.4	(22)
	5%	DB	63.4	63.9	62.7	59.1	56.7	53.0	52.2	52.5	54.7	55.7	58.1	62.0	(23)
		MCWB	57.9	58.8	57.9	54.8	53.1	49.9	48.6	48.2	49.7	51.0	52.7	56.2	(24)
	10%	DB	61.1	61.7	60.5	57.4	55.1	51.8	50.9	51.2	53.0	53.9	56.1	59.9	(25)
		MCWB	56.5	57.5	56.6	54.0	52.4	49.2	47.9	47.9	49.1	50.0	51.5	55.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	63.5	62.5	58.5	56.7	52.9	51.6	51.7	53.2	54.8	58.3	61.3	(27)
		MCDB	67.3	67.9	65.2	60.8	58.0	54.1	53.0	53.3	56.2	58.0	62.1	66.6	(28)
	2%	WB	61.0	61.1	60.2	56.7	55.0	51.7	50.4	50.2	51.8	53.1	55.5	58.9	(29)
		MCDB	64.5	64.9	63.2	59.0	56.6	53.3	52.3	51.8	54.7	56.2	58.7	62.4	(30)
	5%	WB	58.9	59.4	58.5	55.4	53.6	50.5	49.3	49.3	50.6	51.9	53.9	57.4	(31)
		MCDB	62.3	63.1	61.5	58.0	55.5	52.3	51.3	51.1	53.4	54.9	56.9	61.1	(32)
	10%	WB	57.1	57.8	57.1	54.0	52.4	49.4	48.1	48.3	49.5	50.6	52.4	55.7	(33)
		MCDB	60.2	61.1	60.2	56.8	54.6	51.3	50.1	50.5	52.3	53.3	55.3	59.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	8.6	7.9	6.7	5.6	5.4	5.4	6.0	7.0	7.8	8.5	9.2	(35)
		MCDBR	12.4	11.9	11.2	9.3	7.6	6.7	6.6	7.5	9.4	10.4	11.5	12.4	(36)
	5% WB	MCWBR	7.6	7.8	7.2	6.6	5.5	5.7	5.3	5.3	6.1	7.1	7.7	7.7	(37)
		MCWBR	11.0	10.9	9.9	8.1	6.4	6.1	5.9	6.2	8.1	9.0	10.2	11.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.345	0.334	0.327	0.314	0.307	0.309	0.310	0.314	0.339	0.335	0.337	0.338	(40)
		taud	2.541	2.585	2.604	2.619	2.587	2.498	2.535	2.573	2.497	2.530	2.535	2.561	(41)
	Ebn at Noon		308	302	288	266	239	219	233	261	278	298	309	313	(42)
		Edh at Noon	33	30	26	22	18	17	18	22	29	31	33	33	(43)
(44) All-Sky Solar Radiation	RadAvg		1754	1466	1046	661	374	279	341	569	908	1342	1662	1854	(44)
	RadStd		127	83	65	46	23	22	23	36	45	81	120	97	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	+1.08	N/A	N/A	N/A	-95	-238	+137	N/A	

Nomenclature: See separate page